

BODEN, SwedenWMO: **021710**Lat: **65.815N**Lon: **21.640E**Elev: **25**StdP: **101.03**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.6	-26.2	-32.9	0.2	-29.5	-29.2	0.3	-26.2	7.3	0.4	6.2	0.0	0.9	280	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	26.1	18.4	24.2	17.1	22.2	16.0	19.5	24.1	18.2	22.5	17.0	20.9	2.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.8	21.6	16.5	11.8	20.3	15.3	10.9	18.9	55.7	24.0	51.5	22.3	48.0	20.9	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.4	4.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-32.0	21.0	2.8	1.7	-34.1	22.2	-35.7	23.2	-37.3	24.2	-39.4	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-11.4	-9.5	-4.2	1.7	8.1	13.1	16.3	14.0	9.2	2.5	-4.2	-8.6	(6)
(7)		DBStd	10.76	7.57	8.10	5.47	3.58	4.00	3.32	3.01	3.03	3.42	4.41	5.94	7.33	(7)
(8)		HDD10.0	3287	662	547	440	249	85	8	0	4	55	234	426	576	(8)
(9)		HDD18.3	5869	920	781	698	499	317	161	78	138	274	492	676	834	(9)
(10)		CDD10.0	483	0	0	0	0	27	101	197	128	31	1	0	0	(10)
(11)		CDD18.3	22	0	0	0	0	0	3	16	3	0	0	0	0	(11)
(12)		CDH23.3	256	0	0	0	0	15	36	173	32	0	0	0	0	(12)
(13)		CDH26.7	37	0	0	0	0	1	5	28	2	0	0	0	0	(13)
(14)	Wind	WSAvg	2.0	1.7	1.9	2.3	2.1	2.4	2.4	2.1	2.0	2.0	1.7	1.8	(14)	
(15)	Precipitation	PrecAvg	590	46	32	29	31	38	61	72	71	58	50	53	44	(15)
(16)		PrecMax	868	109	58	60	75	117	114	135	155	193	134	99	122	(16)
(17)		PrecMin	400	2	3	10	9	8	8	14	18	11	11	16	8	(17)
(18)		PrecStd	121	22	16	15	16	22	26	31	35	38	26	26	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	7.8	9.3	17.3	25.1	27.1	29.2	26.8	20.2	13.6	9.0	5.0	(19)
(20)			MCWB	4.6	4.7	5.0	9.7	14.5	18.5	20.0	20.1	15.3	9.5	6.1	3.3	(20)
(21)		2%	DB	3.6	5.6	7.2	12.6	22.2	23.7	27.0	23.9	17.8	11.4	7.1	3.1	(21)
(22)			MCWB	1.8	2.9	3.7	7.2	14.0	16.0	18.7	17.6	13.4	8.6	5.8	1.8	(22)
(23)		5%	DB	1.1	3.3	5.7	10.5	18.8	21.8	25.3	21.7	16.2	10.0	5.3	2.1	(23)
(24)			MCWB	0.1	1.4	2.5	5.7	12.2	14.9	18.1	16.7	12.6	8.1	4.0	1.1	(24)
(25)		10%	DB	-0.7	0.9	3.8	8.4	15.9	20.0	23.3	19.8	14.8	8.6	3.5	0.8	(25)
(26)			MCWB	-1.4	-0.2	1.1	4.4	9.8	13.8	17.2	15.6	12.2	6.9	2.7	0.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	5.1	5.5	9.9	15.6	19.3	21.4	20.9	15.7	10.3	6.9	3.6	(27)
(28)			MCDWB	6.5	7.7	8.7	16.5	23.1	26.1	27.2	25.2	18.3	11.8	7.8	4.9	(28)
(29)		2%	WB	2.1	3.2	3.9	7.5	14.3	17.0	20.0	18.8	14.4	9.4	5.8	2.0	(29)
(30)			MCDWB	3.4	5.3	6.6	11.8	21.1	22.0	24.7	22.5	16.6	10.9	7.2	2.7	(30)
(31)		5%	WB	0.3	1.4	2.7	6.0	12.4	15.7	19.0	17.5	13.5	8.2	4.2	1.2	(31)
(32)			MCDWB	1.0	3.0	5.4	9.7	18.1	20.3	23.5	20.6	15.2	9.7	5.2	2.0	(32)
(33)		10%	WB	-1.5	0.1	1.2	4.7	10.6	14.7	17.9	16.2	12.7	7.1	2.7	0.1	(33)
(34)			MCDWB	-0.7	0.9	3.3	8.2	14.9	19.1	22.2	18.8	14.3	8.5	3.5	0.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	8.4	9.4	9.4	10.1	9.2	9.6	9.1	8.2	6.2	5.5	6.8	(35)
(36)			MCDBR	8.1	9.0	10.0	12.9	15.4	12.2	13.4	12.0	10.6	8.2	6.3	6.7	(36)
(37)			MCWBR	7.3	7.2	7.1	7.8	8.1	5.8	6.5	6.7	6.8	6.0	5.4	6.1	(37)
(38)		5% WB	MCDBR	7.9	8.2	9.9	12.3	14.7	10.9	11.4	10.7	8.9	6.9	6.4	6.6	(38)
(39)			MCWBR	7.2	6.8	7.2	7.7	8.0	5.7	6.1	6.6	6.3	5.2	5.5	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	N/A	0.268	0.300	0.332	0.341	0.345	0.359	0.344	0.314	0.292	0.338	0.351	(40)	
(41)		taud	N/A	2.154	2.236	2.331	2.461	2.485	2.465	2.496	2.558	2.397	2.651	2.675	(41)	
(42)		Ebn at Noon	N/A	598	765	833	864	867	841	819	771	587	262	89	(42)	
(43)		Edh at Noon	N/A	54	81	95	93	93	93	81	61	43	18	9	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.10	0.68	2.21	4.03	4.94	5.36	4.98	3.69	2.09	0.78	0.17	0.02	(44)	
(45)		RadStd	0.01	0.10	0.18	0.23	0.49	0.43	0.47	0.34	0.29	0.14	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page